

Fig. S1: Confusion matrices showing the 20 largest histology classes of lung cancer (**top**) and breast cancer (**bottom**). The **left** panel shows the results of a baseline MTCNN classifier while the **right** panel shows the results of our MTCNN DAC classifier. The figures are color coded based on the counts on a log scale: blacks and darker blues indicating higher counts.

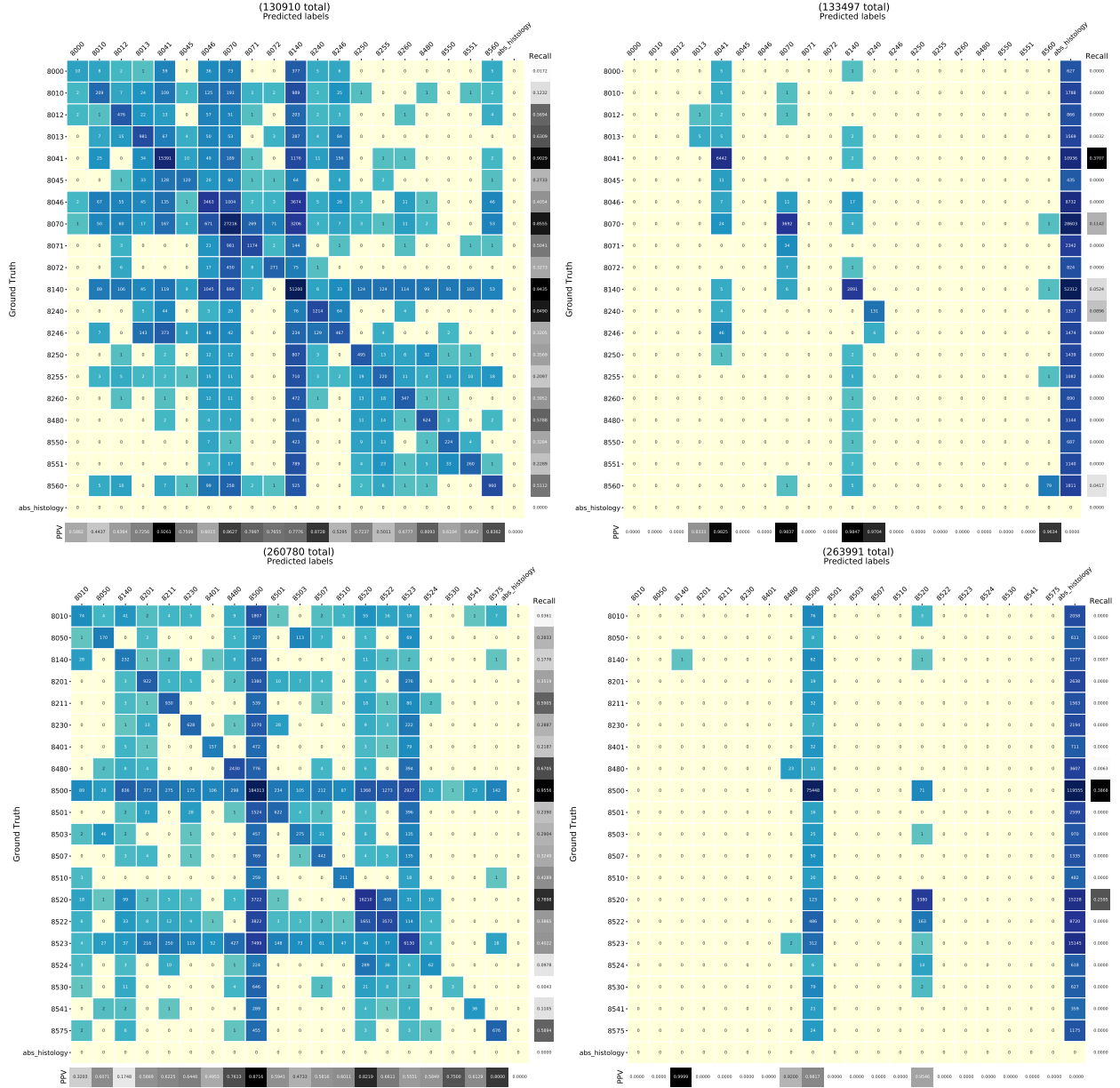


TABLE S1: Number of samples in individual histology classes before and after 3% random noise ingestion; 6 different classes for each lung and breast cancers are shown respectively. The randomization of classes is allowed only within a given topology (site).

Lung cancer		Breast cancer	
No noise	3% noise	No noise	3% noise
8070: 7214	8070: 6998	8500: 76886	8500: 74208
8041: 7073	8041: 6861	8520: 2826	8520: 3269
8140: 6627	8140: 6422	8200: 13	8200: 558
8240: 205	8240: 2285	9020: 4	9020: 561
8013: 103	8013: 188	8480: 2	8480: 550
....			
9699: 1	9699: 68	9699: 1	9699: 586

Local explainability with LIME

In previous work [1] and preliminary analysis for this work, we found that local interpretable model-agnostic explanations (LIME) [2] technique provided plausible local explanation and a useful visual depiction of the explanation. By a *local* explanation, we mean a characterization of what is salient for the classification of a particular sample according to a given task.

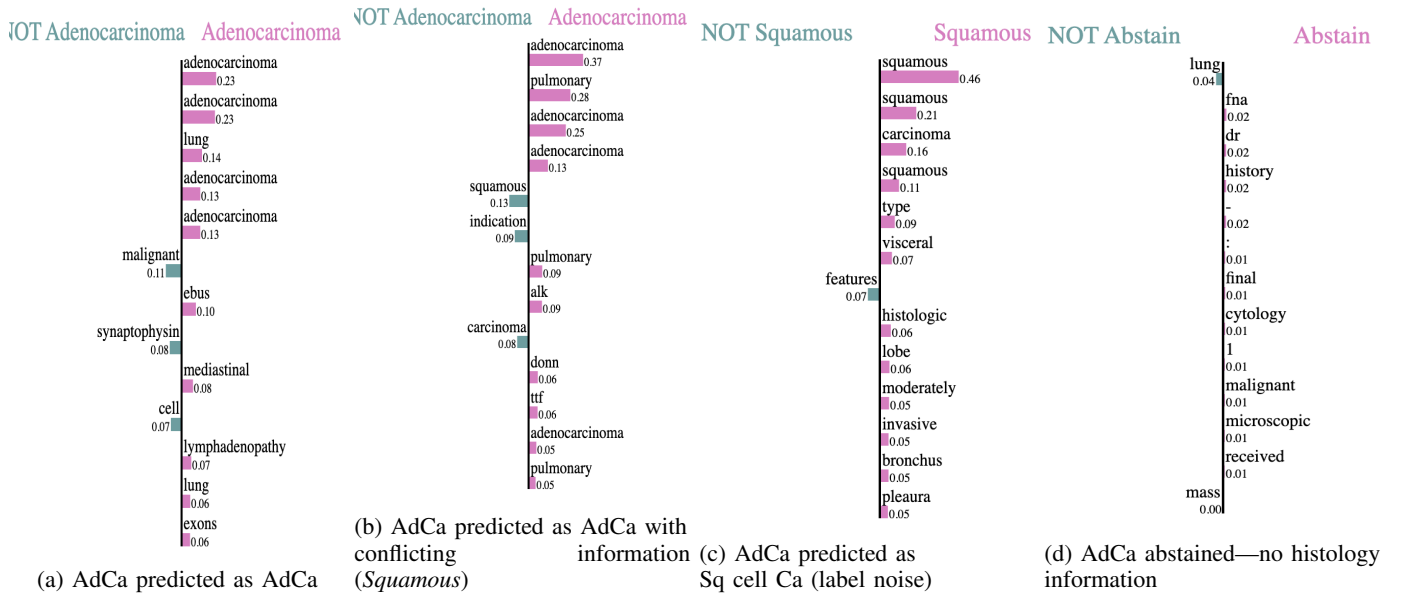
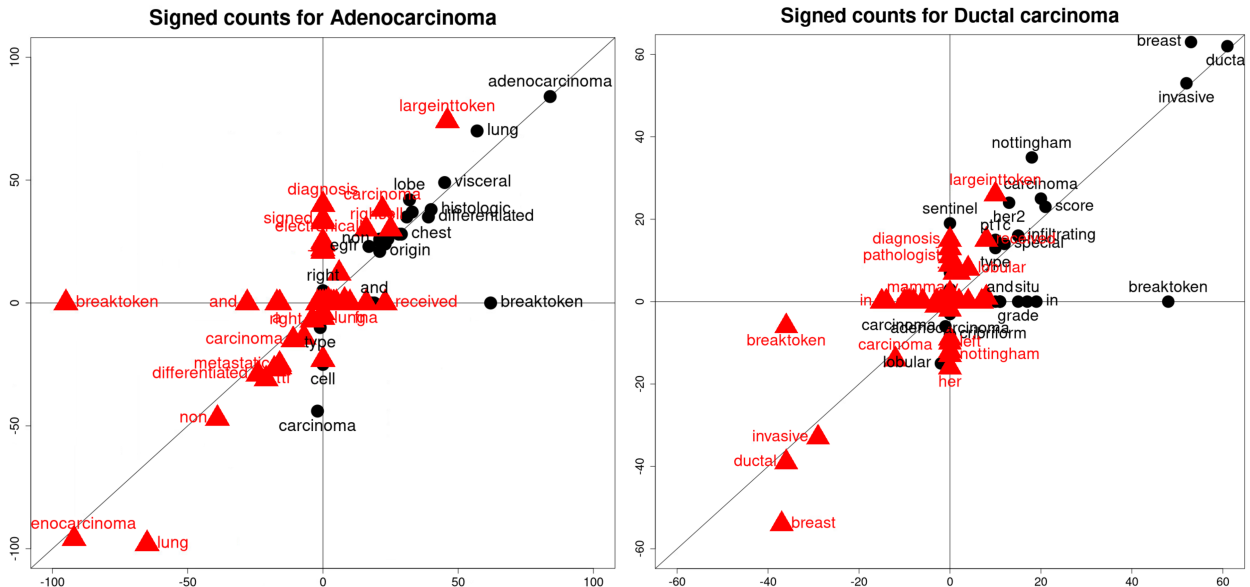


Fig. S2: *Gradient • input (GradInp)* results for four different adenocarcinoma reports highlighting different flavors of confusion. Panels present bar charts of the top 13 (out of 20) words influencing predictions for lung cancer adenocarcinoma cases, including correct and incorrect classifications. Rightward (magenta) bars indicate evidence supporting the predicted class (positive *GradInp* weight), while leftward (olive) bars show evidence opposing it (negative *GradInp* weight). The importance of a word varies depending on its context within the report. Note the clustering of reports by histology type and the influence of conflicting evidence in misclassified and abstained cases.

In contrast, a global explanation would characterize the patterns among these explanations that hold true for a large, possibly the entire, corpus of samples. Though limited to discovering which words-in-context were most important in determining (pro or con) the final prediction class, it still required extensive sampling (50,000 perturbations for each input sample)—and consequently several minutes of computational time per report—to provide repeatable results. In particular, we provide local interpretable model-agnostic explanations (LIME) with a trained multi-task convolutional neural network (MTCNN) deep abstaining classifier (DAC) model and raw pathology reports, prompting it to return the top 40 words-in-context relevant to identifying the winning class. We then increased the default of 5000 input perturbations from which to infer the local linear landscape till multiple runs tended to converge on the same explanations.

Fig. S3: Comparison of *GradInp* explanations (y-axis) with LIME explanations (x-axis) for lung cancer adenocarcinoma and (left) breast cancer ductal carcinoma (right). The axes show the number of reports with report-level positive or negative *GradInp* and LIME weights for the given word in the same set of reports. Black dots show explanations for non-abstained reports and red triangles show explanations for abstained ones



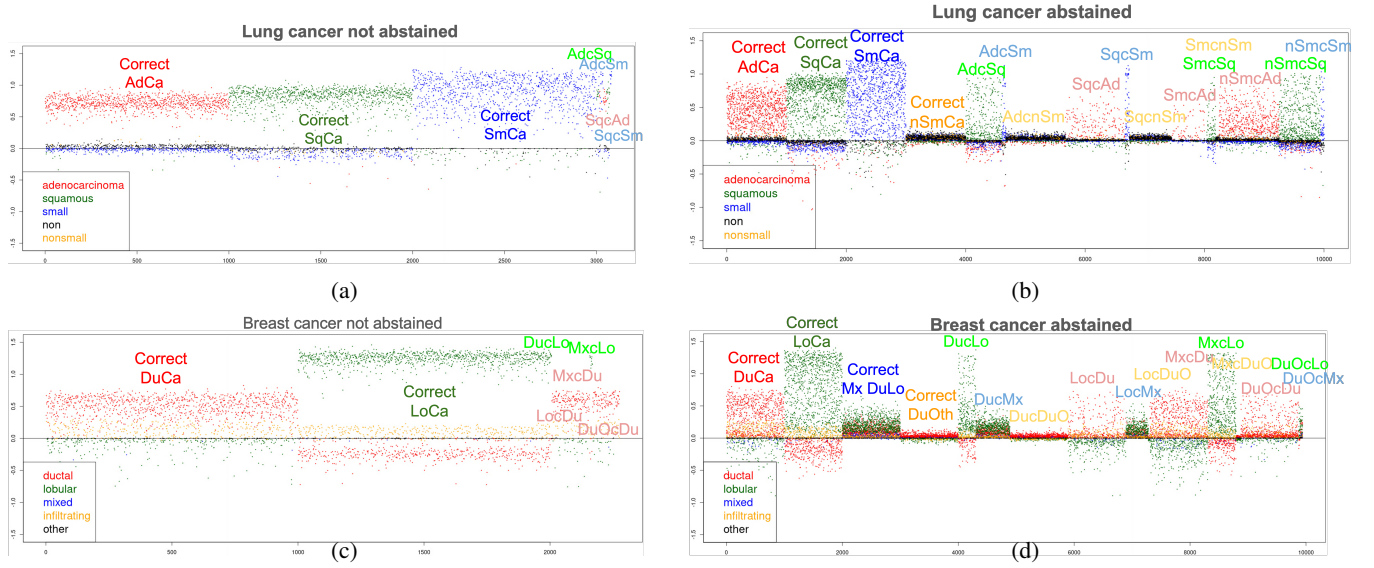


Fig. S4: Local explainability weights distribution in **non-abstained** (left) and **abstained** (right) for four largest histology classes of **lung cancer** (top) and **breast cancer** (bottom) reports. Correct classifications are shown as “Correct <class>” while incorrect classifications are coded as “<ground.truth>c<predicted>”.

DISTRIBUTION OF LOCAL EXPLANATIONS

Figure S4 show importance scores for the most-important words across $\sim 13,000$ reports representing the correct and incorrect classifications in the four largest histology classes of lung and breast cancer respectively. We plot such distributions for both non-abstained as well as abstained samples. The importance scores are computed with respect to the highest softmax scores for the non-abstained samples. For the abstained ones, the importance scores are computed with respect to the second highest softmax scores i.e. the model’s preferred choice in the absence of abstention. Comparison of the distribution of the non-abstained (top) and abstained (bottom) samples shows much higher level of uncertainty in the abstained samples. Important words for correctly classified non-abstained samples appear to be in uniform distribution and deterministic with relatively large positive importance scores for the class-specific words and sometimes corresponding negative scores for words associated with a different class. The distribution of the important words for incorrectly classified non-abstained samples shows more interesting trends, with a high positive score for words associated with the predicted class and sometimes a negative score associated with the ground truth class, suggesting that these samples either have information only on the predicted class or conflicting information between the predicted and ground truth class. Plots of the importance score distribution for the second choice in the abstained samples (lower panel) look much noisier, although following similar pattern as the non-abstained ones. The potential correct classifications have lower positive weights for the class-specific important words and often corresponding negative weights for keywords belonging to different classes, suggesting higher uncertainty. For abstained samples, the large overlap in the distribution between the potentially correct and incorrect classifications, also supports the model’s reasoning for abstaining on those samples.

TABLE S2: Eigenvector matrices for the top 10 words based on magnitude along PC1 and PC2 for **lung cancer** (left) and **breast cancer** (right); variance explained by each principal component are provided in parentheses. The eigenvalues for the first 5 principal components: 0.1249, 0.0960, 0.0484, 0.0059, 0.0032 and 0.2002, 0.0478, 0.0053, 0.0035, 0.0017 for lung and breast cancers, respectively.

	PC1 (39.14%)	PC2 (30.08%)	PC3 (15.17%)	PC4 (1.85%)	PC5 (1.01%)		PC1 (69.75%)	PC2 (16.65%)	PC3 (1.86%)	PC4 (1.23%)	PC5 (0.58%)
small	-0.8625	-0.2034	0.2566	-0.0252	-0.0476	lobular	0.9554	0.1829	0.1885	-0.02786	-0.0323
squamous	0.3971	-0.7892	0.3861	-0.0620	-0.0071	ductal	-0.2665	0.8170	0.4647	-0.07489	-0.0382
adenocarcinoma	0.1312	0.4882	0.8057	-0.0540	-0.1543	breast	0.0443	0.4455	-0.7092	-0.1118	0.1228
carcinoma	-0.1509	-0.2284	0.0826	-0.0182	0.0930	invasive	0.0235	0.2174	-0.1569	0.6936	-0.0452
cell	-0.2083	-0.1566	0.0825	-0.0101	-0.0026	carcinoma	0.0485	0.0713	-0.0645	-0.0290	-0.0654
pulmonary	0.0137	0.0733	0.1348	-0.6205	0.5283	left	0.0512	0.0571	-0.1630	-0.0055	0.3937
bronchial	-0.0123	-0.0527	0.0389	-0.1566	-0.5748	infiltrating	0.0097	0.0720	-0.1087	-0.6794	-0.0282
consistent	-0.0523	-0.0110	0.0445	-0.0087	-0.0271	score	0.0088	0.0628	-0.0576	0.0716	-0.0570
differentiated	0.0350	-0.0277	0.0345	0.0203	-0.0242	nottingham	0.0050	0.0567	-0.0576	0.0960	-0.0594
metastatic	-0.0440	0.0063	0.0523	-0.0138	0.4156	right	0.0276	0.0467	-0.1179	-0.0039	-0.0554

Adaptive Abstention

To better understand how the α parameter is tuned, we describe the actual training of a DAC. The training is carried out by minimizing the DAC loss function while simultaneously tuning α to satisfy two complementary goals: (i) to achieve a target minimum accuracy while (ii) minimizing the total abstention.

Algorithm 1 Algorithm for Alpha Adaptation in Classification

Require: $r_{\max\text{-abs}} \in (0, 1)$, $A_{\min\text{-acc}} \in (0, 1)$ and $\alpha^{(k)} > 0$

Ensure: $\alpha^{(k+1)} > 0$

$e_A \leftarrow \max\{A_{\min\text{-acc}} - A, 0\}$

$e_r \leftarrow \max\{r - r_{\max\text{-abs}}, 0\}$

$a \leftarrow 1 - g_A e_A + g_r e_r$

$\alpha^{(k+1)} \leftarrow a \alpha^{(k)}$

Algorithm 1 describes the process for tuning α , a process that is executed after each training epoch. Let the abstention rate be defined as $r = \# \text{ abstaining samples} / \# \text{ total samples}$, a fraction between 0 and 1. Also, let $r_{\max\text{-abs}}$ denote the target maximum abstention rate and let $A_{\min\text{-acc}}$ denote the target minimum accuracy. These target values are set at the beginning to reflect performance goals and are kept fixed during training. Note that the updating of a , the α adaptive multiplier, depends on e_A , the difference between target and achieved minimum accuracy, and on e_r , the difference between target and achieved abstention rate. A positive e_A means that the accuracy is less than the target value set. As shown in the a update in algorithm 1, this nudges a to be less than one, which in turn means a smaller α , making it cheaper to abstain. This increases the abstention rate and potentially increases the accuracy by removing noisy samples from having any effect on the first term of the DAC loss function. On the other hand, a positive e_r means that the abstention rate is greater than the target set. In the a update, this pushes a to be greater than one, which increments α , making it more expensive to abstain. This should decrease the abstention rate, albeit at the price of a possible decrement in accuracy.

Starting from an initial value of alpha $\alpha^{(0)} > 0$, this adaptive process should achieve a stable α value asymptotically. To avoid large α swings while training, the factor a can be confined to vary in a specified range (e.g. $a \in [0.8, 1.25]$). The adaptation can be made more sensitive to one or the other target setting by using different multiplier factors (i.e. gains g_A, g_r) for e_A and e_r when updating a . Note that we use a validation set during training and base the α adaptation on the metrics computed over this set (not the training set). By this adaptive procedure, it has been shown [3] that the resulting DAC trained models are more robust to feature noise, while providing a powerful tool for identifying uncertain predictions at inference time.

The DAC has also shown to be efficient for multi-task problems [3], i.e. for simultaneously solving multiple non-overlapping classification tasks. For the multi-task case, the DAC poses each subclassification problem as a classification plus abstention, each requiring its own α and its own tuning during training. Earlier versions of the DAC were tuned using a combination of accuracy and abstention targets for each task. In the latest versions, used for this work, we have implemented training methods that, for each task, allow targets for either accuracy or abstention alone, in addition to the original mixed targets. Effectively this is equivalent to setting either $g_A = 0$ (pure abstention target) or $g_r = 0$ (pure accuracy target), but with a modified definition of e_A (respectively, e_r) aiming to preserve accuracy (or abstention rate) inside a given interval, i.e., for accuracy, if it is smaller than the lower limit, it triggers a decrease in alpha, while, if accuracy is greater than the upper limit, it triggers an increase in alpha. For this study, we train only for accuracy; specifically, in order to guarantee 97% accuracy, we choose accuracy targets of $97.5\% \pm 0.5\%$. It is possible to set tighter bounds on the accuracy (or abstention) targets, but this can produce significantly longer training times, especially in a multi-task setting. In a multi-task setting, the stopping criteria require that every task satisfy the desired minimum accuracy target (which can be set independently for each task), except when a task is able to exceed the target with zero abstention, in which case that task is considered to have satisfied its target.

- [1] S. Dhaubhadel, J. Mohd-Yusof, K. Ganguly, G. Chennupati, S. Thulasidasan *et al.*, “Why I’m not answering: Understanding determinants of classification of an abstaining classifier for cancer pathology reports,” *Arxiv*, 2022.
- [2] M. T. Ribeiro, S. Singh, and C. Guestrin, “Why Should I Trust You?: Explaining the Predictions of Any Classifier,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, ser. KDD ’16. New York, NY, USA: Association for Computing Machinery, 2016, p. 1135–1144. [Online]. Available: <https://doi.org/10.1145/2939672.2939778>
- [3] S. Thulasidasan, T. Bhattacharya, J. Bilmes, G. Chennupati, and J. Mohd-Yusof, “Combating label noise in deep learning using abstention,” in *Proceedings of the 36th International Conference on Machine Learning*, K. Chaudhuri and R. Salakhutdinov, Eds., vol. 97, 2019, pp. 6234–6243. [Online]. Available: <http://proceedings.mlr.press/v97/thulasidasan19a.html>